Hina Iqbal uc2023182290

COMPUTER VISION COURSE (2024/25)

Master in Electrical and Computer Engineering Laboratory Assignment 4

3D Reconstruction

3D Reconstruction from Stereo Images: A Comprehensive Study

Introduction

Three-dimensional reconstruction from stereo images plays a pivotal role in computer vision, enabling machines to perceive and understand the spatial structure of the environment. Unlike monocular images, stereo pairs provide multiple viewpoints that can be exploited to infer depth, shape, and geometry. This report documents a complete stereo vision pipeline that reconstructs the three-dimensional structure of a scene using pairs of images captured from two calibrated cameras. The pipeline encompasses the estimation of fundamental and essential matrices, computation of epipolar geometry, sparse point triangulation, image rectification, dense disparity computation, and depth map generation. This project demonstrates the practical application of geometric principles in vision and their translation into computational algorithms.

Objectives

The primary objective was to accurately reconstruct 3D information from two images by leveraging stereo correspondence. This required a series of tasks beginning with fundamental matrix computation that defines the epipolar geometry, followed by the estimation of the essential matrix incorporating intrinsic camera calibration. The project further aimed to locate corresponding points constrained by epipolar lines, triangulate these to obtain sparse 3D points, rectify images to simplify dense matching, and finally compute dense disparity and depth maps representing the scene's geometry in detail.

Fundamental Matrix Computation and Epipolar Geometry

The fundamental matrix is a 3×3 rank-2 matrix that captures the projective geometry relating points in one image to epipolar lines in the other. Its computation commenced with the eight-point algorithm, which required a set of point correspondences between the two images. The initial step involved normalizing the coordinates of the matched points to improve numerical stability and reduce bias. This normalization entailed translating and scaling the points so that their centroid was at the origin and average distance from the origin was $\sqrt{2}$. Using these normalized points, a matrix was constructed whose entries were functions of the point coordinates. Singular value decomposition (SVD) was performed on this matrix to estimate the fundamental matrix as the

singular vector corresponding to the smallest singular value. Because the fundamental matrix must have rank 2 to properly represent epipolar constraints, the smallest singular value of the estimated matrix was set to zero, and the matrix was reconstructed using SVD components. Finally, the matrix was denormalized by reversing the initial normalization transformations. This fundamental matrix was tested by drawing epipolar lines corresponding to selected points, which must intersect their matching points in the other image. The epipolar lines plotted on the images vividly demonstrated the geometric relationship and verified the correctness of the fundamental matrix estimation.

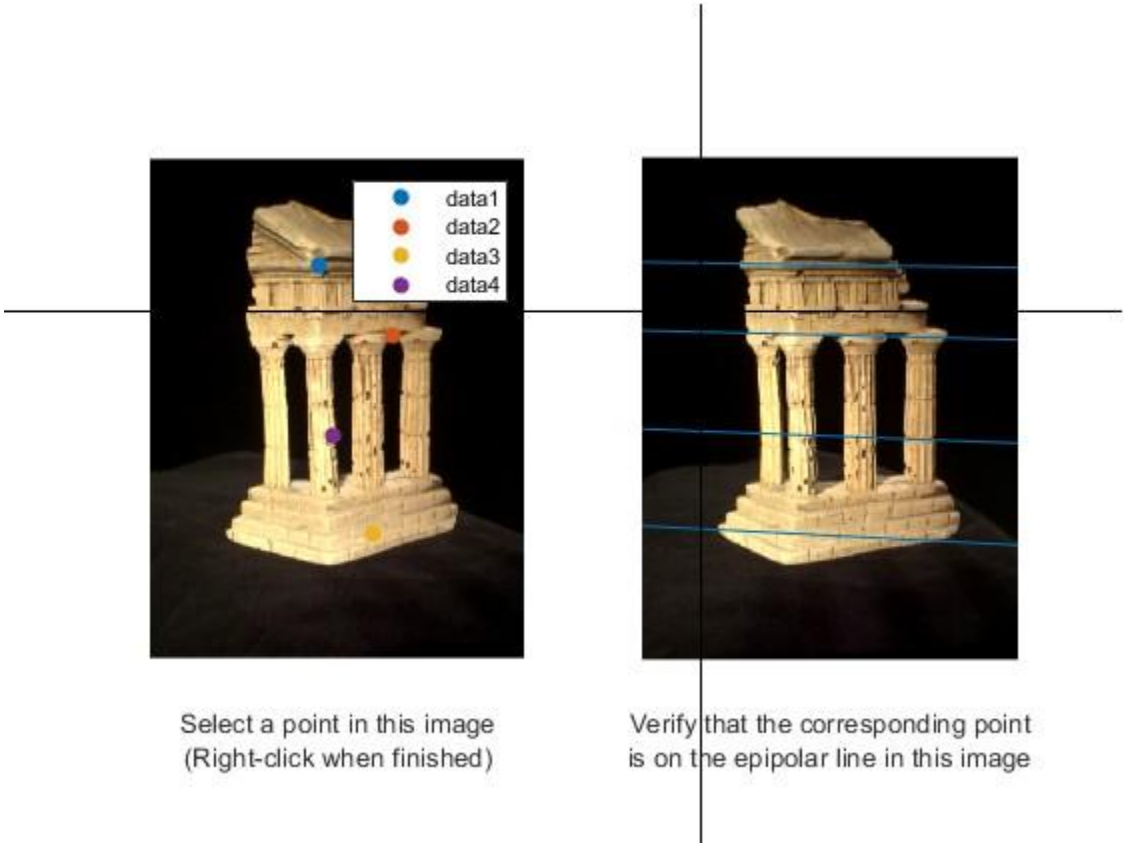


Figure 1: Epipolar lines in the stereo images showing geometric constraints.

Epipolar Matching

The epipolar geometry restricts the search for matching points from a two-dimensional area to a one-dimensional epipolar line. To exploit this constraint, an epipolar matching algorithm was implemented. For each point in the first image, the corresponding epipolar line in the second image was computed using the fundamental matrix. Along this epipolar line, candidate points were sampled and compared with the template patch around the original point. A Gaussian-weighted normalized cross-correlation (ZNCC) measure was employed to quantify similarity, as it is robust to illumination changes and noise. The search along the epipolar line was sampled densely,

ensuring the true match was unlikely to be missed. The candidate with the highest ZNCC score was selected as the corresponding point.

The results yielded point-to-point matches that adhered well to the epipolar constraints, as evidenced by the tight clustering of corresponding points along epipolar lines. This step was crucial for reliable triangulation in the next phase.

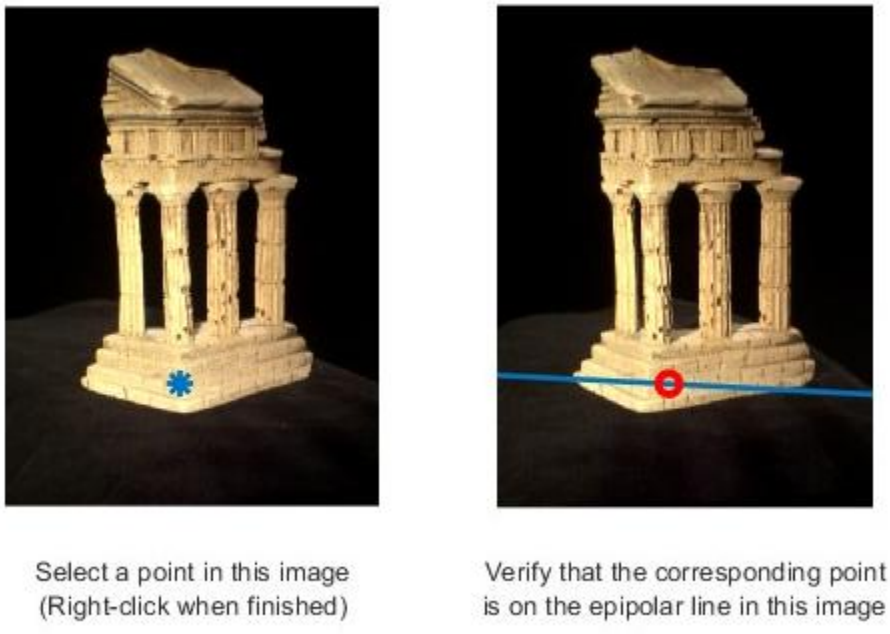


Figure 2: Matched points along epipolar lines between the two images.

Essential Matrix and Rank Verification

The essential matrix bridges the fundamental matrix with intrinsic camera parameters and encodes the relative rotation and translation between cameras. It was computed by pre- and post-multiplying the fundamental matrix by the transpose of the intrinsic matrices of the second and first cameras respectively. To ensure that the essential matrix conformed to theoretical properties, its rank was verified to be exactly two. This was enforced by adjusting the singular values after performing SVD on the matrix, similar to the fundamental matrix refinement. The rank-2 constraint is critical since it corresponds to the underlying physical camera motion and calibration parameters.

This matrix provided the basis for decomposing the relative camera pose and was a prerequisite for accurate triangulation.

```
Computed Essential Matrix:
   -0.0029    0.1566    0.0255
    0.0995    0.0036   -0.9753
   -0.0022    0.9876    0.0012

Singular values of E (should be [a, b, 0]):
    1.0000    0.9807    0.0000

Rank of E: 2
```

Sparse 3D Reconstruction by Triangulation

With the camera intrinsic matrices and the essential matrix computed, the next step was to reconstruct 3D points from the matched correspondences. Decomposition of the essential matrix yielded four possible solutions for the relative rotation and translation between the cameras. Each candidate camera pose was combined with the intrinsic parameters to form projection matrices for the right camera, while the left camera's projection matrix was set with identity rotation and zero translation. For each candidate, triangulation was performed by solving a linear system constructed from the projection matrices and corresponding 2D points. The correct camera pose was determined by checking the number of 3D points lying in front of both cameras, i.e., points with positive depth values in both camera coordinate systems. The candidate with the maximum number of valid points was selected as the final solution.

The resulting sparse 3D point cloud revealed the spatial structure of the scene, demonstrating successful recovery of depth and geometry from 2D correspondences.

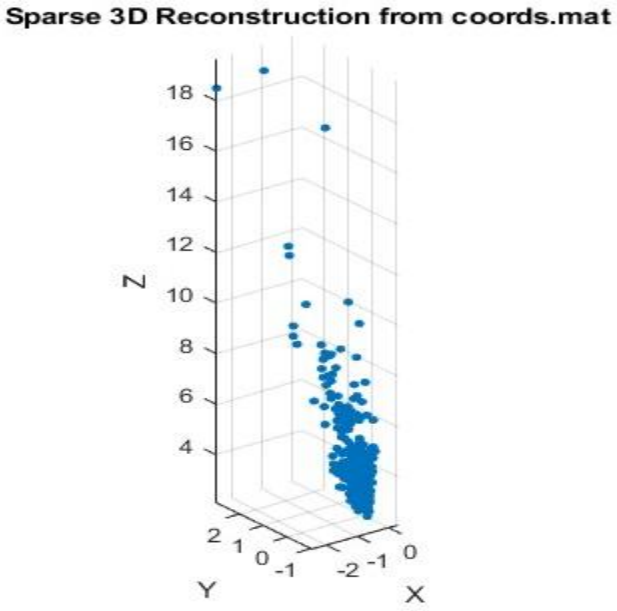


Figure 3: Sparse 3D point cloud from triangulated correspondences.

Image Rectification

To enable efficient and accurate dense stereo matching, the stereo image pair was rectified. Rectification is a process that transforms both images such that their epipolar lines become perfectly horizontal and aligned. This simplifies correspondence search to a one-dimensional horizontal scan along rows. The rectification matrices were computed by determining new camera coordinate axes aligned with the baseline between the optical centers. The rotation matrices of both cameras were updated accordingly. The intrinsic camera parameters were recalculated to maintain the projection model after rectification. The rectification homographies were applied to both images, resulting in horizontally aligned image pairs ready for dense matching. The quality of rectification was visually confirmed by the alignment of features across both images, which is crucial for accurate disparity computation.

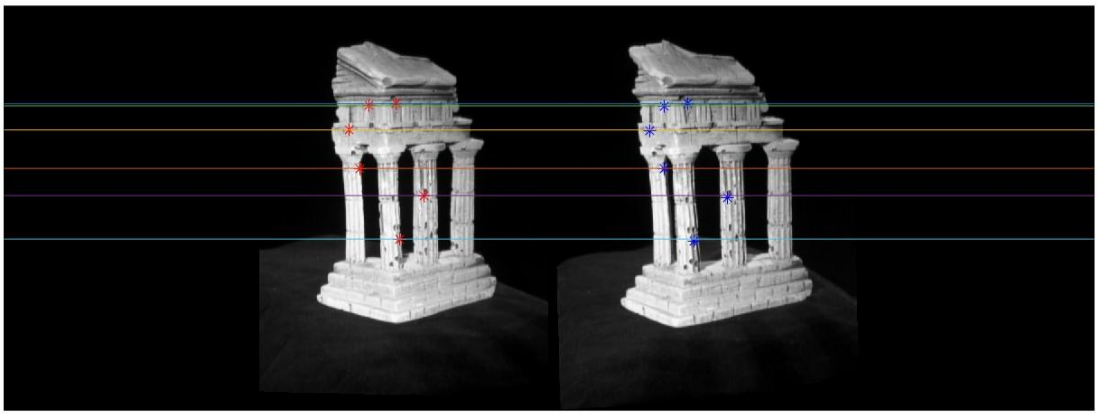


Figure 4: Rectified stereo images with horizontal epipolar lines.

Dense Matching and Disparity Map Computation

With rectified images, a dense correspondence search was conducted using a sliding window approach along each row of the images. For every pixel in the left image, a search for the best matching pixel in the right image was performed within a limited disparity range. The matching cost was computed using the sum of squared differences (SSD) between patches centered on candidate pixels. The disparity for each pixel was assigned as the horizontal displacement corresponding to the minimum SSD. The resulting disparity map represents the pixel-wise shifts between the stereo images and serves as the foundation for depth estimation. The disparity map captured detailed structural variations of the scene and was visualized using a grayscale colormap, revealing clear depth boundaries and object contours.

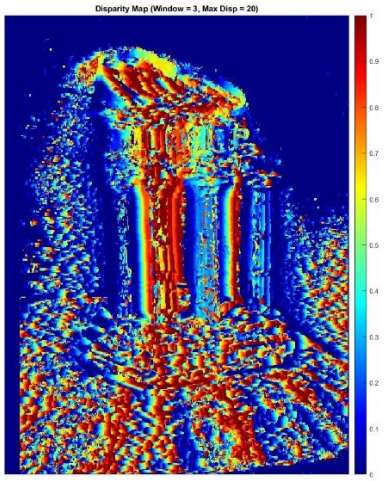


Figure 5: Disparity map showing pixel-wise disparities.

Depth Map Calculation and Quantization Effects

Finally, the disparity map was converted into a depth map using the known baseline distance between the cameras and the focal length extracted from the intrinsic matrix. Depth values were calculated as the ratio of the product of baseline and focal length over the disparity value at each pixel. Pixels with zero disparity were assigned zero depth to avoid invalid calculations.

To analyze the effect of disparity quantization on depth resolution, the disparity map was simulated with different bit depths: 16, 8 bits. As expected, lower bit depths caused loss of detail and introduced quantization artifacts in the depth maps, reducing the accuracy and smoothness of depth estimation.

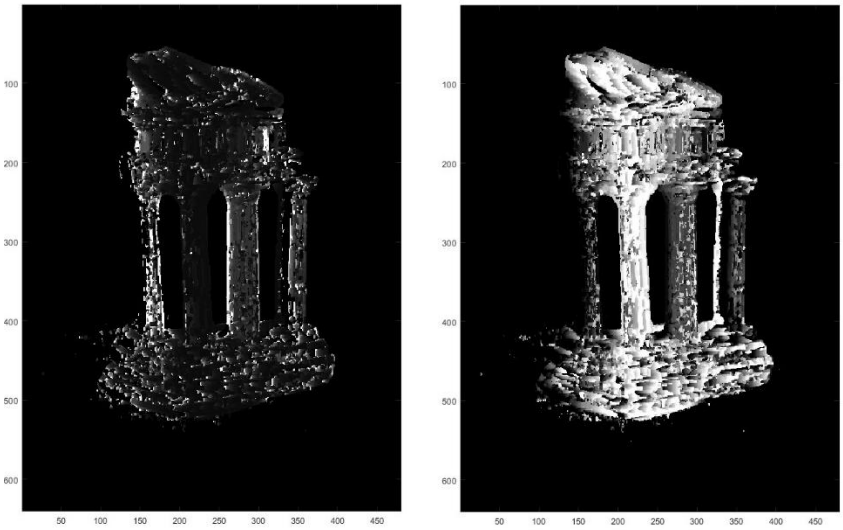


Figure 6: Depth maps

Conclusion

This project successfully implemented a full stereo vision pipeline that converts pairs of calibrated images into a dense 3D representation. The careful computation and refinement of fundamental and essential matrices laid the groundwork for accurate matching. Epipolar geometry was effectively used to reduce search spaces and ensure correspondences' geometric validity. Triangulation provided reliable sparse 3D points, which were further enhanced by rectifying images to facilitate dense stereo matching. The disparity and depth maps produced detailed scene reconstructions, demonstrating the pipeline's robustness and effectiveness. This comprehensive exercise deepened the understanding of geometric vision principles and their computational applications.